Final Project Report

MATH840 Time Series, Final Project. Code: GitHub

Task and models

I forecast the hourly generation of an energy company portfolio for 48 hours (2026-10-27 01:00 to 2026-10-29 00:00) and give a 95% interval for every hour. The history has 33,480 hourly values (2023-01-01 01:00 to 2026-10-27 00:00). The energy mix and units are not given, so I do not call the series solar.

I submitted two files. The final model is the second one.

#	Model	Point forecast
1	ETS Box–Cox 30d + seasonal naive	50% ETS + 50% seasonal naive
2 (final)	Daylight MLP ensemble	25% each: ETS 30d, seasonal naive, daylight MLP seed 1, daylight MLP seed 2

The benchmark is seasonal naive, which repeats the last 24 hours twice. All numbers come from the outputs of the two notebooks in the repository.

1. What I found in the data and how it shaped the model

Data checks. There are no missing values or duplicates, the step is one hour and y is positive everywhere (minimum 0.7), so Box–Cox can be used.

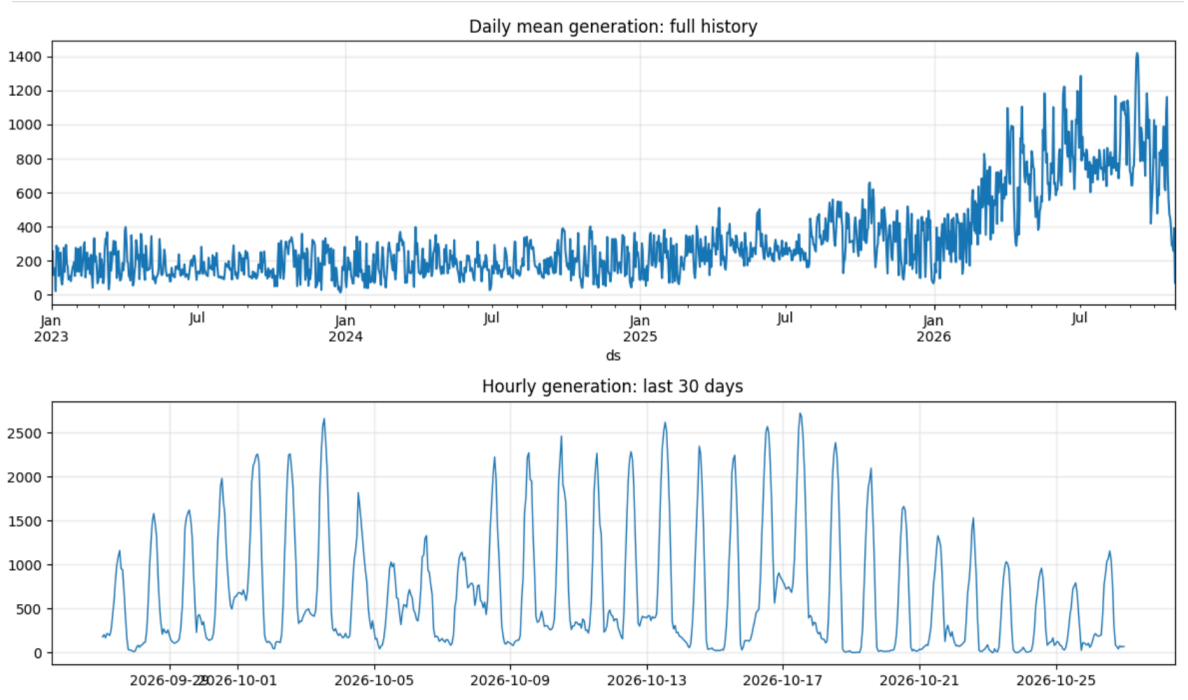


Figure 1. Daily mean generation for the full history (top) and hourly generation for the last 30 days (bottom)

Level changes. The level grows a lot, mostly in 2026.

Year	Hours	Mean y
2023	8,759	170.8
2024	8,784	183.9
2025	8,760	291.5
2026 (until 27 Oct)	7,177	683.8

The level is almost flat in 2023–2024, starts to grow in 2025 and jumps in 2026. Old years look different from the recent series, so I train the models on recent data only (30 to 90 days).

Trend. The recent level rises and falls over several days rather than following a steady trend.

Seasonality. There is a strong daily cycle, so I use a seasonal period of 24 hours. Average generation over the last 90 days is about 1353 during daylight and 308 at night. A separate weekly pattern was not established.

Transformation. Larger levels come with larger daily swings. Box–Cox reduces this difference, so I test it before fitting ETS. Its parameter is estimated separately in each training window. Short-term dependence remains after decomposition.

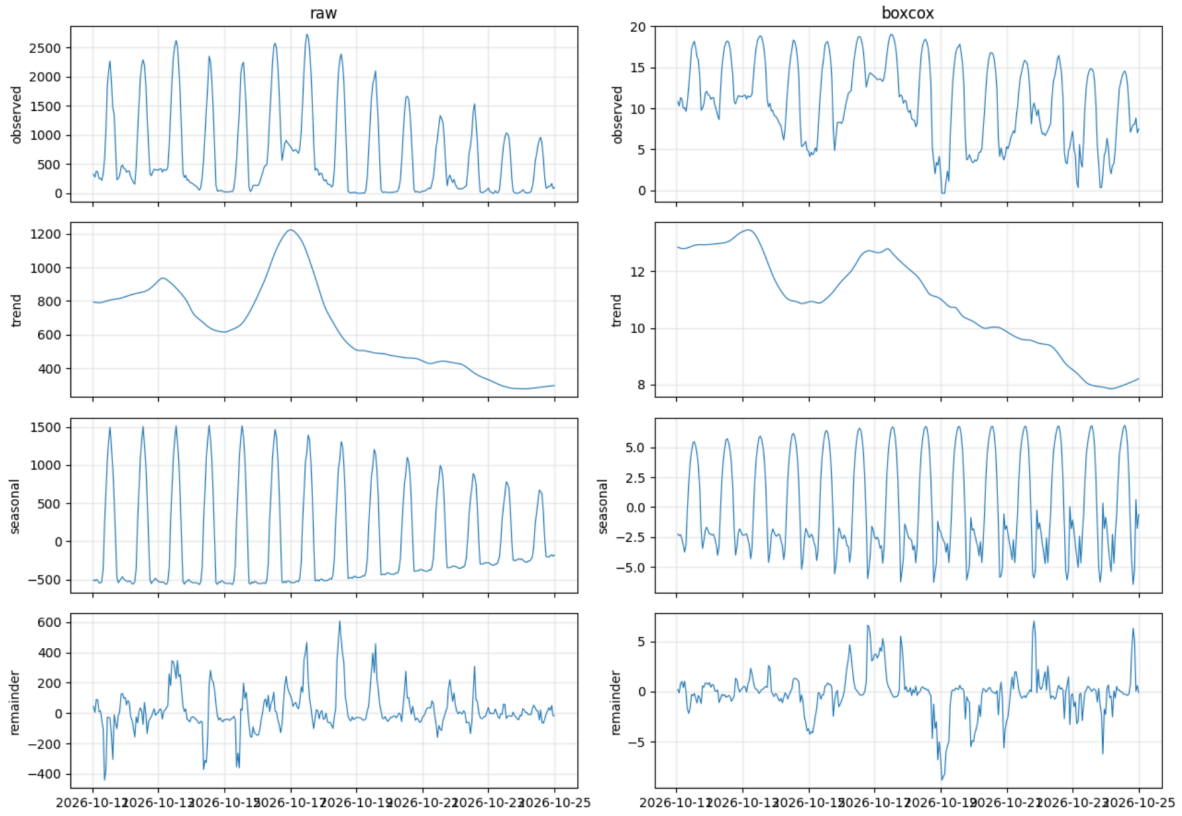


Figure 2. STL decomposition with period 24, last 14 days shown, raw (left) and Box–Cox (right).

The daily peaks repeat every day. On the raw scale their height changes with the level and after Box–Cox the seasonal part is more stable. The trend changes over days, not steadily.

Weather. Generation is associated with daylight, humidity, wind speed and temperature (Appendix A). These associations do not show whether uncertain future weather improves

forecasts.

How the findings changed the model.

Finding	Decision
Strong daily cycle	m = 24 in ETS and seasonal naive; daily Fourier terms in the MLP
Level changes over time	ETS on the last 30 days, MLP on the last 90 days
Swing grows with the level, skewed values	Box–Cox, λ re-estimated in every window
Neighbouring hours depend on each other	MLP input of 168 hours; ETS mixed with seasonal naive
Daylight is the strongest weather link	daylight inputs in the final MLP

Did I use the weather forecast? No. I use `weather_forecast.csv` only for the future timestamps. Historical issued weather forecasts are unavailable and the early tests using repeated-day weather did not show a convincing gain. This does not prove that weather is unhelpful.

The only weather-type input in the final model is `daylight` (day or night). I take its last 168 hours from the training data and repeat the last observed 24 hours twice for the forecast hours.

2. How I checked the forecast

Setup.

- Every window is 48 hours, windows do not overlap, and each forecast uses only data before its start. ETS, Box–Cox λ and the MLP are refitted in every window.
- First search for the ETS model: three search windows and one confirmation window (2026-10-19 to 2026-10-27).
- Main backtest ("recent period"): 28 windows before 2026-10-27 00:00. The first 8 only collect errors for interval calibration, and I score the other 20 (960 hours, from 2026-09-17 01:00).
- Older period: 20 windows from 2026-07-23 01:00 to 2026-09-01 00:00. The first 8 are for calibration and I score 12 (576 hours, from 2026-08-08 01:00).
- I compare with seasonal naive using pooled RMSE (one RMSE over all hours), MAPE, Winkler and coverage. I show day 2 (hours 25–48) separately, because this part is scored on the private leaderboard.
- I used these periods while choosing the models, so the results are retrospective evidence, not an untouched test.

Step 1. ETS alone was not reliable. ETS Box–Cox 90d beat seasonal naive on the three search windows, but lost on the confirmation window (RMSE 197.158 vs 120.404). One window is not enough, so I mixed ETS with seasonal naive.

Step 2. Mix of ETS and seasonal naive, ETS history of 30, 60 or 90 days (20 scored windows, all 48 hours; seasonal naive has its own standard interval).

Model	Pooled RMSE	MAPE	Winkler	Coverage
Combo, ETS 30d	292.316	226.735	1307.860	0.950
Combo, ETS 60d	297.796	205.488	1307.418	0.945
Combo, ETS 90d	301.813	204.574	1286.650	0.935
Seasonal naive	322.016	249.001	2396.705	0.749

The 30-day version has about 9% lower RMSE than seasonal naive and is better than the 90-day version in 14 of 20 windows. I chose 30 days for RMSE. It is a trade-off: 60 and 90 days have lower MAPE, and 90 days has a lower Winkler.

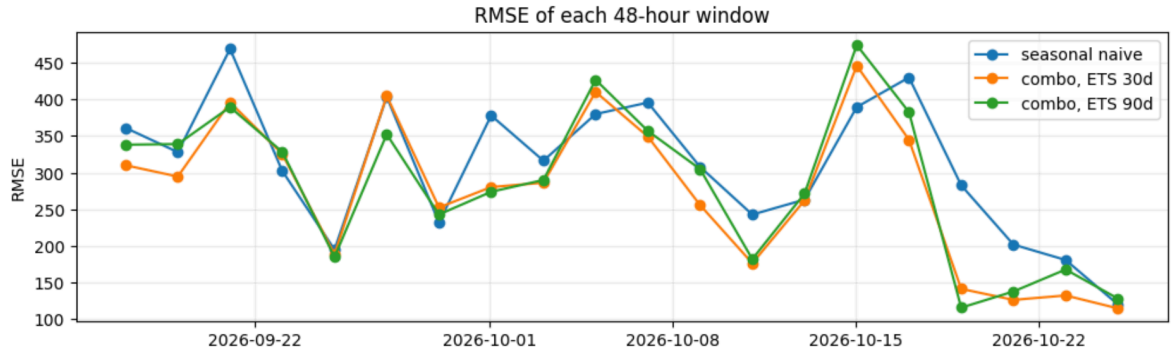


Figure 3. RMSE of each scored 48-hour window

The combination is usually not worse than seasonal naive, but the error changes a lot between windows (for the 30d combo from about 115 to about 445, read from the plot). This is why I compare models on many windows.

Step 3. Final Daylight ensemble, day 2 only. The first model is the 30d combo from Step 2 (same intervals).

Period	Model	Pooled RMSE	MAPE	Winkler	Coverage
Recent (20 windows)	Seasonal naive	334.310	187.639	2095.180	0.823
Recent	First model	306.921	155.302	1401.586	0.948
Recent	Daylight ensemble	298.815	150.405	1244.109	0.960
Older (12 windows)	Seasonal naive	235.346	65.226	1151.088	0.941
Older	First model	211.713	59.561	1146.474	0.951
Older	Daylight ensemble	206.592	51.199	1040.078	0.986

On day 2 the Daylight ensemble has about 11% lower RMSE than seasonal naive in the recent period and about 12% lower in the older period. Compared with the first model, the gain is only 2–3%.

Compared with the same ensemble without daylight, daylight improves all three metrics recently. On the older period, it slightly worsens RMSE but improves MAPE and Winkler. The benefit is small and not uniform.

The ensemble does not win every comparison. Over all 48 recent forecast hours, the first model has slightly lower pooled RMSE (292.316 vs 293.621; Appendix B). Averaging window RMSE can also give a different ranking. I did not test statistical significance.

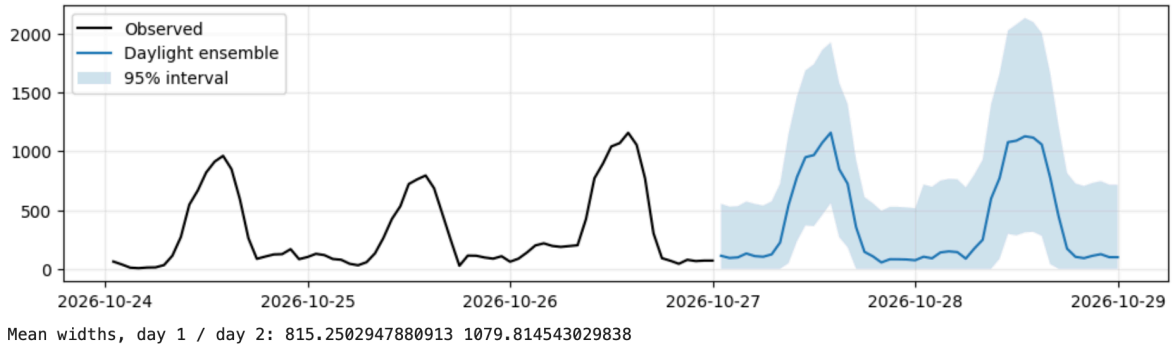
3. How I built the 95% intervals

Method for the final ensemble.

1. Each component gives its own 95% bounds. ETS uses its own bounds, transformed back from the Box–Cox scale. Seasonal naive and the MLP use $\hat{y} \pm 1.96 \cdot \sigma \cdot \sqrt{(1+k)}$, where σ is the standard deviation of y(t) − y(t−24) in the training window and k is 0 for day 1 and 1 for day 2. For the MLP this is a simple rule, not a learned distribution.
2. I average the bounds with the same weights as the forecasts (25% each). This is a practical recipe, not exact quantiles of the mixture.
3. I widen the interval using the last 12 completed 48-hour forecasts. For each hour I measure how far the actual value was outside its interval (0 if inside) and divide it by the half-width of that interval. I take the 95% quantile separately for day 1 and day 2 and multiply it by the new interval's half-width. I subtract this adjustment from the lower bound and add it to the upper bound. The lower bound is clipped at zero.

Effect of the widening (first model with ETS 90d, 20 windows, all 48 hours): coverage rose from 0.843 to 0.935 and Winkler fell from 1466.129 to 1286.650, although the mean width grew from 837.651 to 1053.443.

Results for the final model. The coverage and Winkler are in the Step 3 table. In the recent period coverage is 0.960, close to the 95% target. In the older period it is 0.986, so the intervals are conservative. Winkler is better than seasonal naive in both periods. A wide interval costs Winkler even when coverage is high, and the widening step can only widen. Because of this, intervals can be too wide on quiet days. The method does not guarantee 95% coverage for a series with a changing level.




```
Mean widths, day 1 / day 2: 815.2502947880913 1079.814543029838
```

Figure 4. Final forecast with the 95% interval

The interval is wide at the daily peaks and narrow at night. It is wider on day 2: the mean width is about 815 on day 1 and about 1080 on day 2.

4. Diagnostics and what did not work

Forecast errors. I checked the final ensemble's errors on the 20 scored recent windows (960 hours), using a time plot, a histogram and ACF. Error means actual minus forecast, so negative bias means the forecast is too high.

The model often makes errors in the same direction for several hours. The ACF also has a smaller peak around lag 24, suggesting some daily dependence remains.

Forecast day	Hours	Bias	RMSE	MAPE (%)
Day 1, daylight	207	−38.310	381.039	32.413
Day 1, night	273	−1.799	189.956	499.995
Day 2, daylight	206	−64.058	378.943	34.275
Day 2, night	274	−13.317	220.139	237.714

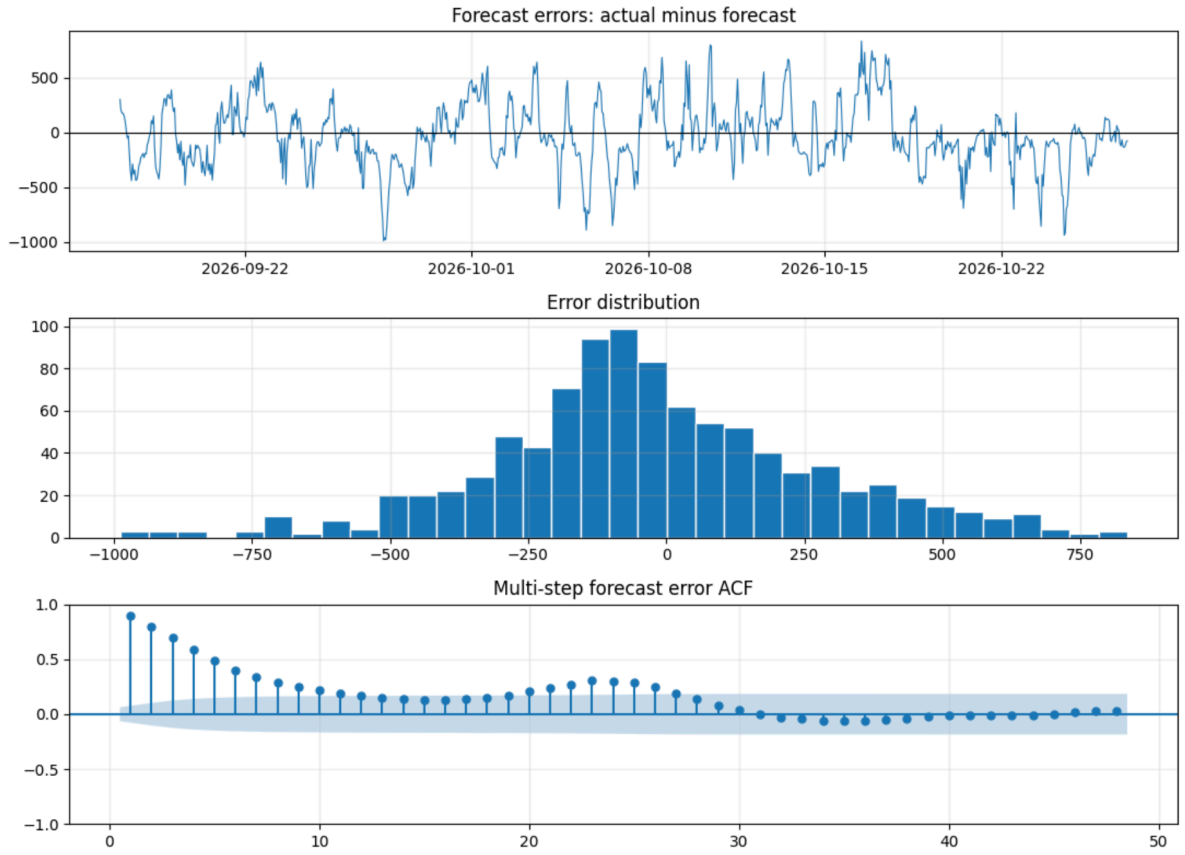


Figure 6. Validation forecast errors: time plot, distribution and ACF over the 20 scored recent windows

Errors are larger during daylight. On average, the daytime forecast is too high by about 38 units on day 1 and 64 on day 2. At night, absolute errors are smaller, but MAPE is high because actual generation can be close to zero.

These are multi-step validation errors, not one-step model residuals. Their ACF is descriptive; I do not claim that the fitted ensemble's residuals are white noise. This check did not change the submitted model.

I also inspected one difficult forecast window below.

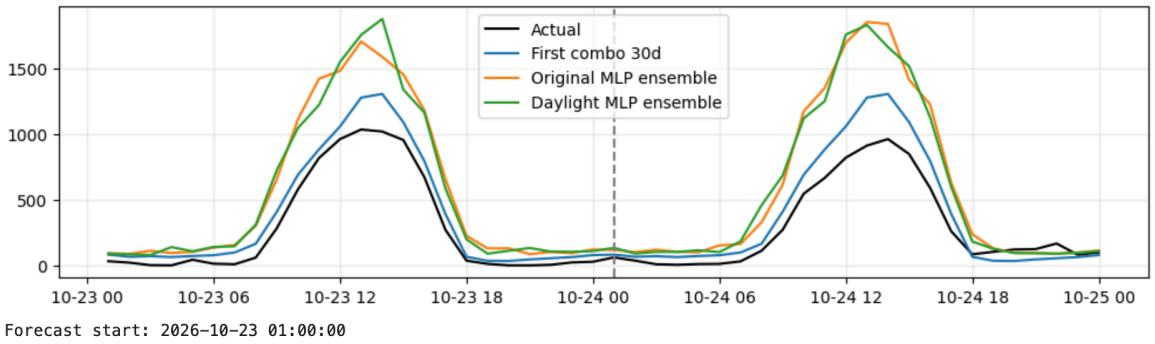


Figure 5. The window where the original MLP ensemble increases day-2 RMSE most compared with the first model (forecast start 2026-10-23 01:00)

Both MLP ensembles overpredict the peaks: about 1700–1900 versus about 1000 observed. The first model overpredicts less. Averaging seeds cannot remove a shared error. This window was selected for its large deterioration, so it is not representative; I did not tune a peak cap to it.

Not explained.

- Why the level grew in 2025–2026. The energy mix and units are not given, so I cannot explain it. Short training windows only help the models follow it.
- Why both MLPs overpredict the same peaks.

Candidates that did not give a clear gain. These were tested on different periods, so this is not one common ranking.

Candidate	Result	Decision
ETS Box–Cox alone	Lost on the confirmation window (197.158 vs 120.404)	Mixed with seasonal naive
MLP alone	Pooled RMSE 347.324 vs 292.316 for the first model (recent, all 48 hours)	Used only in the ensemble
MLP with one seed	Pooled RMSE 298.937 and 303.139 vs 296.947 for two seeds (recent, all 48 hours)	Averaged two seeds
Compact MLP inputs	RMSE slightly better, MAPE and recent Winkler worse	Kept the original inputs
Boosting, Prophet, STL + ETS, regression with ARIMA errors	No convincing gain in the early comparisons	Not used
Ridge + first model	Better on recent windows, small gain on the older period	Kept as an alternative only
NeuralForecast MLP and N-BEATS	Only short runs, too little to judge	No conclusion about these models

Additional SARIMA experiments. I also tried simple seasonal ARIMA models with 7 days of training data and a daily period of 24 hours. Different settings worked better on different periods, with no consistent winner. I did not complete a comparison with the Daylight ensemble on identical forecast dates before the deadline, so I kept the submitted ensemble.

Conclusions

- The series has a strong daily cycle and an unstable level, so I use short training windows, Box–Cox and a mix with seasonal naive.
- On the backtest windows, the final Daylight ensemble has about 11–12% lower day 2 pooled RMSE than seasonal naive and only 2–3% lower than the first model, and not in every comparison.
- The intervals cover about 96% in the recent period and about 99% in the older period, and they have better Winkler than seasonal naive.
- The private result (hours 25–48) is not known yet, so I make no claim about beating the benchmark.

Appendix A. Correlation of y with observed weather

Variable	Full history	Last 90 days
daylight	0.395	0.721
wind_speed	0.303	0.561
temperature	0.280	0.506
cloud_cover	0.162	0.151
pressure	−0.025	0.050
air_density	−0.260	−0.455
humidity	−0.380	−0.597

Appendix B. All 48 hours (same windows and intervals as in Step 3)

Period	Model	Pooled RMSE	MAPE	Winkler	Coverage
Recent (20 windows)	Seasonal naive	322.016	249.001	2396.705	0.749
Recent	First model	292.316	226.735	1307.860	0.950
Recent	Daylight ensemble	293.621	224.378	1273.946	0.957
Older (12 windows)	Seasonal naive	209.199	57.894	1035.117	0.910
Older	First model	205.985	54.124	1241.721	0.955
Older	Daylight ensemble	204.219	49.074	978.364	0.977

Appendix C. Model details

- ETS: additive errors, damped additive trend, additive daily seasonality, fitted on Box–Cox of the last 30 days. I transform back without a bias adjustment, so the forecast is a median-type value.

- Seasonal naive: the last 24 hours repeated twice.
- MLP: direct 48-hour output, trained on the last 90 days. Box–Cox in the training window, StandardScaler for inputs and targets, two hidden layers (64 and 32), alpha = 1, at most 100 iterations, seeds 1 and 2.
- MLP inputs: the last 168 transformed values; mean and standard deviation over 24, 72 and 168 hours; 5 daily Fourier pairs and a weekend flag for the forecast hours; daylight as described in Section 1.